

The Hydra-Centaurus Supercluster

An exotic collection of galaxies awaits observers willing to tackle the southern sky. | **By Steve Gottlieb**

WILLIAM HERSCHEL, POSSIBLY history's greatest deep-sky observer, began a systematic visual survey of the heavens in late-18th-century England with an 18.7-inch reflector. His son, John, completed this ambitious all-sky survey when he took his father's telescope to Africa's Cape of Good Hope and mapped the southern skies during the 1830s. Sir John's discoveries included a rich collection of "nebulae," now known as the Hydra-Centaurus Supercluster, which spans the constellations Antlia, Hydra, and Centaurus.

Both Herschels employed scientific methods to classify, measure, and catalog the hazy patches of light they discovered, but they had only a rudimentary comprehension of the structure of the universe. Throughout the years their diligent efforts have served as inspiration as astronomers developed more sophisticated equipment and gathered more reliable data. By the 1970s, large-scale redshift surveys allowed astronomers to stitch together detailed three-dimensional maps showing the distribution of galaxies in the local universe.

One feature of the universe to emerge from this data is superclusters of galaxies — enormous flattened or filamentary chains of galaxy clusters stretching more than 100 million light-years. Although the organization of superclusters is still being unraveled, they form a frothy cellular network consisting of chains and walls surrounding equally huge voids.

Equipped with sharper (and sometimes larger) optics than the early visual pioneers and aided by a better understanding of what we are seeing, today's amateurs are fortunate to be able to gaze upon these immense structures.

After the Virgo Supercluster (also known as the Local Supercluster), the

Hydra-Centaurus complex is the nearest supercluster, and its relative proximity yields several easily accessible fields for intrepid galaxy hunters. This month we

can tour the individual galaxy clusters Abell 3526, 3565, and 3574 and along the way encounter a cosmic zoo of exotic galaxies.

At declination -41° , the Centaurus Cluster (**Abell 3526**) skirts the southern horizon from my observing site in the Sierra Nevada foothills of Northern Cali-



Located in Centaurus deep in the southern sky, the spiral galaxy NGC 4622 is most unusual. Contrary to what human experience suggests from its appearance in this image obtained with the Hubble Space Telescope, the galaxy is rotating clockwise — the well-defined outer arms are leading the direction of rotation. NGC 4622 is part of the Hydra-Centaurus Supercluster discussed in the accompanying article. Details of the galaxy's unusual rotation are available at <http://bama.ua.edu/~rbuta/ngc4622/pr.html>.

HUBBLE HERITAGE TEAM (AURA / STSC) / NASA

fornia. Although midnorthern observers tend to shy away from galaxies this far south, I find it an exciting challenge to scour the southern constellations using a detailed star atlas or computer-generated charts. Because the cluster lies at a modest distance of about 165 million light-years, a half dozen of its members are magnitude 11½ or brighter.

The core of the Centaurus Cluster is dominated by **NGC 4696**, located 9° northwest of grand globular star cluster Omega Centauri. This 10½-magnitude giant elliptical galaxy is a strong source of X-rays, and recent studies suggest that the nucleus houses a supermassive black hole. My 17½-inch Dobsonian reflector reveals a 2'-by-1½' oval with a brighter core. Look for a 13½-magnitude star touching the northwest edge. Curiously, John Herschel described this galaxy as resolvable — possibly because he had the impression it was mottled.

NGC 4709, the cluster's second-ranked member, lies ¼° east-southeast of NGC 4696. This 11th-magnitude elliptical spans about 1' and is slightly oval in an east-west orientation. **NGC 4706**, a slightly smaller 13th-magnitude companion 6' to the north, is elongated south-southwest to north-northeast and displays a modest central brightening. Herschel mentioned that he found "more nebulae hereabouts," and indeed a cloud of dim galaxies surrounds NGC 4709. Spend some time in this region, and try to snare some of the fainter cluster members.

If you center NGC 4709 in your eyepiece and turn off the telescope's drive motor and let the sky drift across the field of view for two minutes, **NGC 4743**, a 1'-by-½' spindle of light, should float into view. If you nudge your scope ¼° northward you come to **NGC 4729** and **NGC 4730**, a pair of 12½-magnitude glows separated by 3'. Both are ¾' in diameter and weakly concentrated. Due to

imprecise coordinates reported by John Herschel, the identifications of NGC 4729 and NGC 4730 were misapplied in some older references, which can be confusing if you use multiple observing guides. **NGC 4744** is a faint spiral located 7' northeast of the pair.

On the northwest side of the cluster is the Centaurus Chain — 10 galaxies stretched across a 45'-by-11' band of sky. First in the chain is **NGC 4603**, a 2'-diameter face-on spiral of fairly low surface brightness. Heading 24' northeast brings you to **NGC 4622**. Visually you'll find a fairly uniform glow, but NGC 4622 is a bizarre spiral. A recent study using the Hubble Space Telescope demonstrates that this galaxy is rotating "backward," with the two outer spiral arms leading the direction of rotation — a most unusual situation. Less than ¼° east is a small pair of interacting galaxies, **NGC 4622A/B**. Their cores are separated by just 20", so use 200× or more to resolve this duo. Another 6' eastward is a much larger spiral, **NGC 4650**, with a bright bar oriented west-northwest to east-southeast embedded in a round, dim halo.

The chain ends with the exotic polar-ring galaxy **NGC 4650A**, which acquired its odd form when one galaxy experienced a close encounter or merger with a companion. Under excellent conditions, scopes of 25-inch aperture or larger may resolve a remarkable cigar-shaped lenticular galaxy intersected at right angles by a small, edge-on ring consisting of younger stars.

Let's leave the Centaurus Cluster and head 12° northeastward for a short visit to **Abell 3565**, our second stop in the Hydra-Centaurus Supercluster. Abell 3565 is ruled by **IC 4296**, a 10½-magnitude radio galaxy. I noted it as a moderately bright 2'-diameter halo, which appears condensed with a prominent core and stellar nucleus. Look for a smaller spiral



HUBBLE HERITAGE TEAM (AURA / STScI / NASA)

NGC 4650A, located just ½° east of NGC 4622, is another unusual galaxy. One of only about 100 known "polar-ring" galaxies, NGC 4650A's bizarre form is believed to have arisen from the collision of galaxies at least 1 billion years ago. While the inner disk has the yellow color characteristic of a galaxy consisting of old stars, the polar ring is dappled with the blue light of hot young stars, which astronomers think formed in the ring of dust and gas that began orbiting the older galaxy in the collision's aftermath.

companion, **IC 4299**, just 6' to the south-southeast. It measures about 1' across and brightens to a small, round core.

John Herschel missed this pair when he swept through the cluster in 1836, but he found the nearby interacting system **NGC 5215**. At 220×, I was able to resolve two small, 13th-magnitude components separated by roughly 20". Both galaxies exhibit sharp, stellar nuclei. Not surprisingly, Herschel mistook the western component for a faint star.

Just 10' due east, **NGC 5220** is easy to locate northwest of a 9th-magnitude star. This 12th-magnitude edge-on galaxy appears about 1½' long and half as wide

Selected Galaxies in the Hydra-Centaurus Supercluster

Name	R.A. (2000.0)	Dec.	V Mag.	Size	Notes
NGC 4622	12 ^h 42.6 ^m	-40° 45'	12.4	1.7' × 1.6'	Spiral arms leading rotation
NGC 4650	12 ^h 44.3 ^m	-40° 44'	11.6	3.2' × 2.8'	Part of the Centaurus Chain
NGC 4650A	12 ^h 44.8 ^m	-40° 43'	13.3	1.6' × 0.8'	Polar-ring galaxy
NGC 4696	12 ^h 48.8 ^m	-41° 19'	10.4	4.5' × 3.2'	Part of Abell 3526
NGC 4709	12 ^h 50.1 ^m	-41° 23'	10.9	2.4' × 2.0'	Part of Abell 3526
NGC 5215	13 ^h 35.2 ^m	-33° 29'	12.9	1.1' × 0.5'	Interacting pair
NGC 5220	13 ^h 35.9 ^m	-33° 27'	12.0	2.3' × 0.7'	Bisecting dust lane
IC 4296	13 ^h 36.7 ^m	-33° 58'	10.6	2.8' × 2.8'	Part of Abell 3565
IC 4329	13 ^h 49.1 ^m	-30° 18'	11.3	3.4' × 1.9'	Part of Abell 3574
IC 4329A	13 ^h 49.3 ^m	-30° 19'	13.0	1.4' × 0.4'	Seyfert galaxy
NGC 5291	13 ^h 47.4 ^m	-30° 24'	14.1	1.2' × 0.8'	Seashell Galaxy nearby

and is punctuated by a 13th-magnitude star on the eastern side. A strong dust lane similar to the one associated with the famous Sombrero Galaxy (M104) bisects this galaxy. Can anyone catch a glimpse of this seldom-observed feature?

Our final destination is **Abell 3574**, better known as the IC 4329 cluster. The new edition of *Uranometria 2000.0* includes a close-up chart of this group, which plots more than 40 galaxies in the immediate vicinity.

At magnitude 11.3, **IC 4329** should be readily visible in an 8-inch scope from a dark site. In my 17½-inch the moderately bright oval is about 1½' long and contains a bright core and stellar nucleus. A 13th-magnitude companion to its east, **IC 4329A**, is a well-studied Seyfert galaxy with an active nucleus likely powered by a massive black hole. Visually it appears as a small spindle, elongated southwest to northeast with a sharp nucleus.

In the same field 10' to the southwest you'll find **NGC 5298**, a 13th-magnitude oval glow, which is weakly concentrated. About 6' to its west you can look for **MCG-05-33-010** hiding among a group of faint stars. Several other galaxies, including **NGC 5302**, should be visible in the same field with a 12-inch or larger scope.

We'll end our tour at **NGC 5291** along the western edge of Abell 3574. Deep images reveal this unusual system to be a disturbed, gas-rich lenticular galaxy with long plumes containing nearly two dozen H II knots. Visually, the galaxy is just a hazy 14th-magnitude glow, but it's humbling to contemplate the length of its tidal tails extending six times the diameter of our Milky Way galaxy! **MCG-05-33-005**, an interacting neighbor, appears as a small, round knot, nearly attached to the southwest edge of NGC 5291. This strange companion is dubbed the "Sea-shell Galaxy" due to a distorted whelk shape seen in deep photographs.

These three rich clusters have been relatively unexplored by northern observers and should provide exciting challenges for backyard observers with moderate and large scopes and clear southern horizons. Let me know how you fare the next clear April evening.

STEVE GOTTLIEB is a regular *Deep-Sky Notebook* contributor. His last article, "Exploring the Pisces-Perseus Supercluster," appeared in the January 2001 issue. He can be reached by e-mail at sgottlieb@telis.org.